

RESEARCH ARTICLE

Intelligent Spacecraft Attitude Fault Recovery Using Deep Reinforcement Learning

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ABSTRACT Reliable satellite attitude fault recovery is essential for the success of space missions, particularly as satellites increasingly operate autonomously in dynamic and uncertain environments. Such uncertainties may arise from actuator degradations, subsystem anomalies, environmental disturbances, or other unpredictable operational conditions. Maintaining control resilience in these scenarios is critical to preserving mission objectives and ensuring system stability. However, traditional Proportional-Derivative (PD) controllers and existing deep reinforcement learning (DRL) algorithms such as TD3, PPO, and A2C often fall short in providing the real-time adaptability and robustness required for autonomous satellite operations. This study introduces a DRL-based fault recovery strategy designed for satellite resilience and adaptability under uncertainty. In contrast to previous DRL-based solutions that focused primarily on nominal operations or minimal noises, this work introduces a new method tailored for fault recovery. The proposed method, referred to as TD3-HD, employs Twin Delayed Deep Deterministic Policy Gradient (TD3) enhanced with Hindsight Experience Replay (HER) and Dimension-Wise Clipping (DWC) to accelerate learning under sparse post-fault rewards and stabilizes policy updates in degraded scenarios. This combination forms a distinct learning mechanism designed specifically for resilience and adaptability in the presence of actuator faults and structural variations. Experimental results demonstrate that TD3-HD achieves significantly lower attitude error, improved angular velocity regulation, and enhanced stability in the presence of uncertainty, highlighting its potential as a robust fault recovery framework for future autonomous satellite missions.

INDEX TERMS Deep reinforcement learning, twin-delayed deep deterministic policy gradient, hindsight experience replay, attitude control, reaction wheels, spacecraft autonomy.

I. INTRODUCTION

With the growing deployment of small satellites in low Earth orbit (LEO) for Earth observation, communications, science, and defense, there is an increasing demand for intelligent agents that can learn, adapt, and ensure resilience of the spacecraft in uncertain environments [1]. Sustained mission success now relies heavily on adaptability, which serves as the foundation of reliable and long-duration operations.

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Uncertainties in space missions arise from a variety of sources, including environmental effects such as atmospheric drag and solar radiation pressure, subsystem variability like actuator wear and sensor noise, and unexpected anomalies/intrusion in hardware or software [2]. Traditional approaches most notably Proportional-Derivative (PD) controllers perform well under predefined conditions but struggle when dealing with dynamic uncertainties due to their reliance on fixed gains. Their limited adaptability constrains real-time responsiveness, raising the risk of degraded performance, compromised mission objectives, or even mission failure when unforeseen disruptions occur [3].

Subsystem vulnerabilities further highlight the need for adaptive autonomy. Reaction wheels (RWs), for example, may degrade over time, threatening attitude stability, while power and thermal subsystems face similar risks that can affect spacecraft performance. Conventional mitigation often depends on human intervention or rule-based for known faults, thereby limiting autonomy in evolving operational contexts. Addressing these challenges requires intelligent frameworks capable of learning and adapting on their own to preserve functionality under uncertain conditions [4].

Recent advances in artificial intelligence, particularly deep reinforcement learning (DRL), offer promising avenues to enhance autonomy through agents that learn policies directly from interaction with their environment. DRL enables agents to adapt dynamically by refining decisions through experience rather than relying on pre-programmed responses. Trained predominantly in high-fidelity simulations with randomized scenarios, these agents achieve zero-shot generalization to different conditions, making them well suited for improving spacecraft resilience against faults, degradations, and environmental disturbances [5].

As discussed in [6], autonomous intelligent agent requires anomaly/intrusion detection functionality in addition to recovery from fault as well. Cyber-intrusion and cyber-physical attacks are an increasingly recognized threat to control systems, including spacecrafts, and a substantial body of work applies machine learning (including deep learning and reinforcement learning) to intrusion and anomaly detection in industrial control systems (ICSs). Recent surveys summarize ML/DRL approaches for detecting network-level and physical-process anomalies and identify challenges such as data imbalance, concept drift, and deployment constraints in safety-critical systems [7], [8].

Within this context, DRL has emerged as a *promising solution* for intrusion detection in ICSs and supervisory control and data acquisition (SCADA) environments. Unlike traditional supervised learning approaches that rely on static datasets and offline retraining, DRL-based intrusion detection systems formulate detection as a sequential decision-making problem, enabling agents to continuously adapt their policies in response to evolving and previously unseen attack patterns.

This promise has been substantiated by recent studies demonstrating the practical application of DRL to intrusion detection systems in operational technology environments. In particular, [9] work on reinforcement learning based resilience and decision making in cyber physical systems highlights the suitability of RL for security-critical settings characterized by uncertainty, incomplete information, and adversarial behavior.

More concretely, [10] apply multiple DRL algorithms, including DQN, DDQN, Dueling DDQN, PPO, A2C, and REINFORCE, to network intrusion detection in ICS using realistic datasets featuring Modbus-TCP and DNP3 protocols. Their results demonstrate that DRL-based intrusion detection systems can achieve high detection accuracy with

reduced false positive rates while maintaining robustness to dynamic attack behaviors. The study further shows that, through appropriate model adaptation strategies such as mini-batch state generation, simplified reward functions, and discount-factor tuning, DRL-based IDS can be effectively trained and deployed even in the absence of live attack environments.

Related reinforcement learning applications in industrial cyber-physical systems, such as demand response management in smart grids with prosumers [9], further reinforce the role of DRL as a viable and scalable decision-making framework for safety-critical control systems. Collectively, these works establish DRL-based intrusion detection not only as a promising research direction but as a validated and practically applicable solution, motivating its integration into autonomous intelligent agents alongside fault diagnosis and recovery capabilities.

Although our focus is on fault recovery (e.g., actuator failure, sensor faults, power anomalies), several ideas from ICS intrusion detection are directly relevant. For example, (i) telemetry- and network-level anomaly detectors can provide early warning signals that trigger recovery policies; and (ii) DRL-based IDS methods suggest mechanisms for learning adaptive detection strategies against evolving attacks. At the same time, satellites pose domain-specific constraints (intermittent downlink, extreme resource limits, and the requirement for provable safety under autonomy) that require adaptation of ICS techniques before deployment in orbit [11]. Hence, intrusion/anomaly detection methods should be considered as complementary to our recovery agent.

This study introduces TD3-HD, a DRL-based agent derived from the Twin Delayed Deep Deterministic Policy Gradient (TD3) algorithm [12], enhanced with Hindsight Experience Replay (HER) [13] to accelerate learning from sparse feedback and Dimension-Wise Clipping (DWC) [14] to improve policy stability and ensure safety in agent actions. TD3-HD embodies adaptive autonomy: TD3 ensures stable policy refinement, HER converts failed trajectories into informative training samples, and DWC enforces safe adaptation across action dimensions. While spacecraft attitude management using RWs is employed here as a representative case study, the proposed framework is broadly applicable to diverse operational uncertainties. The contributions of this work are threefold:

- Identifying the limitations of conventional and current DRL approaches in enabling adaptive learning for attitude fault recovery and autonomy under subsystem variability and environmental uncertainty.
- Proposing TD3-HD as an enhanced DRL agent designed to accelerate adaptive policy learning and resilience in uncertain spacecraft operations.
- Benchmarking TD3-HD against PPO, A2C, and standard TD3 within the Basilisk simulation framework, demonstrating its superior learning efficiency,

adaptability, and resilience under both nominal and faulty scenarios.

The remainder of the paper is organized as follows: Section II reviews relevant literature on adaptive learning and DRL applications for spacecraft resilience. Section III formulates the problem as a Markov Decision Process (MDP) under uncertainty. Section IV details the TD3-HD framework, including its reward design and adaptive objectives. Section V outlines the Basilisk-based experimental setup. Section VI presents empirical results, followed by discussion in Section VII and concluding remarks in Section 8.

A. ABBREVIATIONS AND ACRONYMS

ADCS	Attitude Determination and Control System.
A2C	Advantage Actor-Critic.
DWC	Dimension-Wise Clipping.
DRL	Deep Reinforcement Learning.
HER	Hindsight Experience Replay.
LEO	Low Earth Orbit.
MDP	Markov Decision Process.
MRP	Modified Rodrigues Parameters.
PD	Proportional-Derivative.
PPO	Proximal Policy Optimization.
RW	Reaction Wheel.
SAC	Soft Actor-Critic.
TD3	Twin Delayed Deep Deterministic Policy Gradient.
TD3-HD	TD3 with Hindsight Experience Replay and DWC.
IS	Importance Sampling.

II. RELATED WORKS

This section outlines the evolution of spacecraft autonomy, starting from classical approaches to modern (RL) agents that emphasize adaptive intelligence and resilience in uncertain operational domains. Conventional methods established the foundation for stability but remain limited in addressing the uncertainties of space, where subsystem degradations and environmental disturbances necessitate continual adaptation to preserve mission objectives. In contrast, advances in DRL have enabled agents to acquire resilient policies through direct interaction with their environment, reducing dependence on pre-programmed rules and fostering real autonomy. Figure 1 illustrates this transition, highlighting the movement toward learning-centered frameworks that prioritize adaptability, safety, and robustness. This review synthesizes significant milestones in RL driven spacecraft autonomy, leveraging insights from high-fidelity simulations and early real-world applications to illuminate pathways for enhanced mission resilience [15], [16].

A. BACKGROUND

A wide range of techniques have been explored in spacecraft operations, from traditional controllers such as Proportional (P) methods to more advanced strategies like sliding mode,

adaptive, fuzzy logic, and neural networks. While these approaches improved robustness and addressed specific challenges such as variability or nonlinearities, they still rely on predefined rules or model-based assumptions. As a result, their capacity to support long-term autonomy under unforeseen conditions remains limited. This motivates a transition toward learning-based agents that can acquire adaptive behaviors directly from experience, enabling greater resilience and independence in uncertain environments.

- **Proportional controllers:** Proportional-Derivative (PD) and Proportional-Integral-Derivative (PID) controllers represent the traditional foundation of spacecraft control, valued for their simplicity and effectiveness in achieving accurate pointing under nominal conditions. Yet, as summarized in Table 1, their performance degrades significantly in the presence of disturbances, varying environments, or parametric uncertainties [17]. This limitation arises from their dependence on fixed gain settings, which restricts adaptability and hinders resilience in the face of nonlinearities and dynamic mission scenarios [18], [19].
- **Sliding Mode Control (SMC):** SMC represents a significant advancement beyond classical PID controllers, offering robustness against disturbances, uncertainties, and actuator faults that threaten system autonomy. As outlined in Table 2, numerous SMC variants have been developed to mitigate the well-known chattering effect while enhancing performance and resilience in spacecraft operations. Adaptive SMC, for example, employs real-time disturbance estimation to increase flexibility and allow dynamic adaptation to evolving conditions [23]. However, persistent chattering and high computational demands limit the practicality of SMC in resource-constrained spacecraft. To address these challenges, several enhanced approaches have been explored. Fuzzy Sliding Mode Control (FSMC) integrates fuzzy logic to reduce chattering and achieve smoother control [24], though at the expense of additional complexity and parameter sensitivity. Minimum Sliding Mode Error Feedback Control (MSMEFC) reduces sliding mode errors for higher precision [25], but increases computational load and tuning difficulty. Adaptive Non-Singular Terminal SMC (ANSTSMC) avoids singularities and ensures finite-time convergence through adaptive tuning [26], yet its complexity hampers real-time feasibility. Adaptive Fuzzy SMC (AFSMC) further combines fuzzy elements with adaptive SMC for agile adjustments and improved tracking [27], while Integral SMC (ISMC) incorporates integral actions to improve steady state accuracy under persistent disturbances [28]. Each of these extensions enhances robustness but introduces additional design or resource burdens.

A notable application was reported in [29], where actuator jitter in micro-satellites caused by reaction

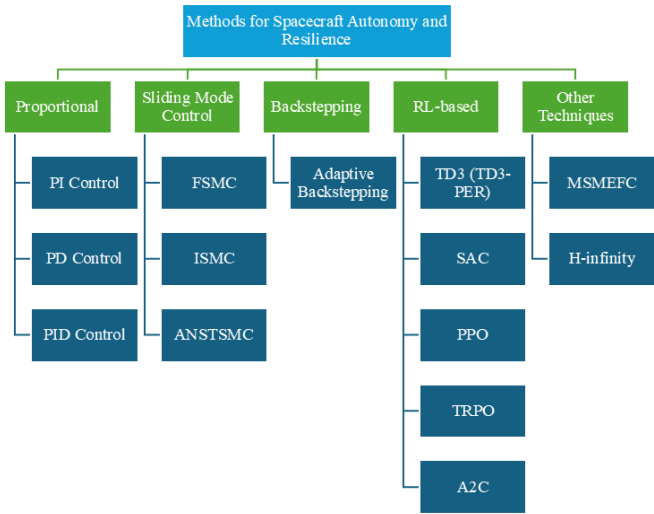


FIGURE 1. Overview of the studied approaches.

TABLE 1. Comparison of proportional controllers.

Method	Advantages	Disadvantages	Ref.
P Control	Simple, low cost, fast response	Cannot eliminate steady-state errors	[20], [21]
PI Control	Eliminates steady-state errors, higher precision	Slow response to rapid changes.	[19], [21]
PD Control	Reduces overshoot, fast response	Cannot handle steady-state errors	[18]
PID Control	Simple structure, widely applied	Requires tuning, not robust to disturbances	[21], [22]

TABLE 2. Comparison of sliding mode control variants.

Method	Advantages	Disadvantages	Ref.
SMC	Robust to disturbances, low computational cost	Causes chattering, actuator wear	[23]
FSMC	Handles uncertainties, reduces chattering	High computational cost	[24]
ISMC	Effective in magnetic attitude control	Cannot fully neutralize disturbances	[28]
ANSTSMC	Fast convergence, strong robustness	Computationally demanding	[26]

wheel vibrations was mitigated using a hybrid SMC with adaptive torque distribution. The SMC component countered bounded disturbances, while the adaptive algorithm redistributed torques based on wheel dynamics, significantly reducing attitude errors to 0.001° compared with PD baselines. Despite these promising results, reliance on simplified models and the need for extensive tuning underscore the trade-offs between robustness and practical deployment. Overall, SMC and its variants substantially improve resilience, precision, and fault tolerance. However, the computational demands and design complexity present barriers for real-time, resource-limited spacecraft, highlighting the need for continued refinement to balance robustness with autonomy and efficiency.

- **Backstepping Controller:** Backstepping was introduced in [30] as a systematic framework for stabilizing non-linear systems by decomposing them into recursive sub-systems. This methodology improves trajectory tracking and enhances robustness, particularly in applications such as space manipulators, where it has been combined with sliding mode control to maintain performance under uncertainties and disturbances. As summarized in Table 3, several Backstepping variants have been developed to address practical challenges. For example, a study applied Backstepping with Modified Rodrigues Parameters (MRP) and an extended state observer for satellite attitude control under disturbances [31]. The approach demonstrated strong disturbance rejection but introduced increased computational

complexity and chattering. An adaptive Backstepping fault-tolerant control (FTC) strategy for actuator faults was proposed in [32], showing effective fault handling but suffering from chattering and heavy computational demands. Backstepping was applied to microsatellites with inertia uncertainties, achieving improved stability [33], though requiring extensive fine-tuning and posing computational difficulties for real-time use in constrained environments.

Overall, Backstepping offers a powerful nonlinear control technique with strong resilience to uncertainties and actuator faults. However, its high computational cost, susceptibility to chattering, and dependence on careful parameter tuning remain significant obstacles to its practical implementation in real-time, resource-limited spacecraft systems.

- H_∞ Controller: H_∞ control was introduced in [34] as a robust framework designed to withstand uncertainties, disturbances, and noise in control systems. It ensures stability even without precise environmental models, making it particularly suitable for complex spacecraft systems where unpredictability is prevalent. By optimizing worst-case disturbance rejection, H_∞ provides spacecraft with enhanced resilience and reliability in challenging operational conditions. As summarized in Table 4, numerous implementations have been proposed to reinforce spacecraft autonomy.

For instance, a study addressed spacecraft attitude control under actuator misalignments and external disturbances using a nonlinear H_∞ controller, achieving stability through the Chebyshev–Galerkin solution of the Hamilton–Jacobi–Isaacs (HJI) equation [35]. While effective, this approach is often mathematically complex and can yield conservative results under nominal conditions. Similarly, another study applied a backstepping based nonlinear H_∞ controller for flexible spacecraft, demonstrating vibration attenuation and finite-time stability during large-angle maneuvers [36]. However, the computational burden and demanding parameter tuning requirements limit its applicability in real-time, resource-constrained scenarios.

In summary, H_∞ control offers significant robustness and fault tolerance for spacecraft operations, but its high complexity, heavy computational overhead, and conservative performance under nominal conditions highlight the need for further research to improve efficiency and adaptability for practical deployment.

Despite the advances offered by the previous techniques, their inherent complexity and computational intensity often limit their effectiveness for real-time applications, particularly in small satellite systems operating under uncertain conditions. These limitations restrict practical deployment in resource constrained missions, where rapid adaptation and efficiency are critical. Consequently, ongoing research is required to refine such methods into solutions that balance

robustness with operational feasibility, ensuring scalability and resilience in practice.

This gap has accelerated interest in reinforcement learning (RL), a paradigm that enables autonomous agents to adapt through experiential policy refinement rather than relying on predefined control laws. RL offers strong potential for advancing spacecraft autonomy and resilience, especially in environments characterized by dynamic uncertainties. The following section therefore surveys the integration of RL into spacecraft systems, highlighting core methodologies, benefits, and open challenges, with illustrative applications such as attitude management using actuators like reaction wheels (RWs).

B. RL FOR SPACECRAFT AUTONOMY AND RESILIENCE

Artificial intelligence (AI) has long been recognized for its potential to address complex real-world problems [37]. Foundational methods such as backpropagation [38] initially showed promise but only gained practical impact once computational power and data availability matured. Early applications of AI in the satellite domain, such as those pursued at INPE [39], were constrained by the technological limitations of their time. With the advancement of machine learning and AI architectures, renewed interest has driven the development of intelligent agents capable of adaptation and resilience.

Reinforcement learning (RL), closely related to optimal control principles [40], provides a framework in which agents learn through interaction with their environment rather than relying on explicit system models. As summarized in Table 5, algorithms such as Soft Actor-Critic (SAC), Proximal Policy Optimization (PPO), Twin Delayed Deep Deterministic Policy Gradient (TD3), and Advantage Actor-Critic (A2C) have been explored for spacecraft applications due to their ability to handle nonlinear dynamics, adapt under uncertainty, and enhance subsystem resilience. The development of Deep Q-Networks (DQN) [41] enabled scalable value function approximation, while the Deep Deterministic Policy Gradient (DDPG) algorithm [42] extended RL to continuous action spaces. Subsequent improvements, including TD3 [12] and SAC [43], addressed overestimation and stability issues, while policy gradient methods such as PPO [44] and TRPO [45] established reliable performance guarantees.

Several studies highlight RL's growing role in spacecraft control. SAC has been applied to predictive control for nonlinear uncertainties [46], showing adaptability for actuators such as thrusters and flywheels, although efficiency was limited by line-search dependencies. RL combined with PD controllers has been employed in docking scenarios [47], though robustness was reduced by sparse rewards and computational overheads. Q-learning has been used for model-free post-capture control [48], demonstrating adaptability but limited resilience under sparse rewards. A2C has been explored for fault-resilient adaptation using a Stackelberg framework [49], achieving real-time adaptability but at high

TABLE 3. Comparison of backstepping-based methods.

Method	Advantages	Disadvantages	Ref.
Backstepping	Handles nonlinearities, Lyapunov stability	Computationally expensive	[31]
Adaptive Backstepping	Adapts to uncertainties, high precision	High complexity, tuning required.	[33]

TABLE 4. Comparison of H_∞ control techniques.

Method	Advantages	Disadvantages	Ref.
H-infinity	High robustness, performance bounds	Complex design, matrix equation solving	[35]
backstepping-based H_∞	Energy-efficient, fault-resilient	Limited maneuver handling.	[36]

computational cost. SAC and PPO based controllers have been demonstrated in Basilisk for debris removal [50], enabling adaptability across varying mass profiles, though outlier scenarios remained challenging.

Other efforts have emphasized sample efficiency and guided learning. Expert-Guided Exploration (EGE) has been introduced to accelerate policy convergence by mimicking PID responses, enhancing anomaly tolerance but increasing computational demands [51]. PPO has been applied for planetary landing trajectories [52], achieving accurate and fuel-efficient control but with high sample requirements and sensitivity to hyper-parameters. PID-assisted TD3 has been proposed to improve adaptation under perturbations and actuator saturation [53], though pretraining demands limit its on-orbit applicability. TD3-PER has also been used on micro-satellites such as Diwata [54], improving sample efficiency and training stability, yet its performance degrades in sparse-reward regimes where prioritized replay over-emphasizes noisy transitions.

Collectively, above studies demonstrate model-based and adaptive fault-tolerant control (FTC) methods are fundamental approaches and provide provable stability guarantees and work well when accurate models and online parameter estimates are available. However, modern satellite fault scenarios often involve complex nonlinearities, partial observability, and unmodeled interactions (thermal, power, and actuator saturation) where deriving and updating high-fidelity models online is costly or infeasible. Recent works indicate that DRL-based controllers can outperform model-based and adaptive alternatives in such settings by learning robust, nonlinear recovery strategies from diverse offline experience and by gracefully handling partial observability and unmodeled disturbances [55], [56]. Therefore, our focus in this paper is on DRL-based approaches. However, these approaches including TD3, still struggles with slow adaptation, limited fault learning, and instability in highly uncertain or faulted environments.

To overcome these challenges, this work introduces TD3-HD a deep reinforcement learning framework that integrates HER and DWC. This combination forms a distinct and new learning mechanism, rather than a mere performance-

oriented modification, designed to improve adaptability and robustness under actuator faults and structural variations. Unlike prior methods that sequentially combine existing components, TD3-HD establishes a novel interaction mechanism between HER and DWC within the actor-critic learning loop. Specifically, in TD3-HD:

- **HER** is not simply appended as an external replay strategy; it dynamically redefines goal-conditioned transitions during critic updates, allowing the agent to reinterpret failed experiences in relation to dimension-wise action constraints imposed by DWC.
- **DWC**, in turn, introduces adaptive, per-dimension policy gradient regulation, which alters how the actor network learns from relabeled HER samples effectively coupling exploration, safety, and credit assignment at the learning-rule level.
- This **bidirectional** coupling between HER and DWC creates a new training dynamic that adjusts gradient flow and target estimation jointly, enabling stable learning under sparse and safety-critical reward signals.

Therefore, TD3-HD represents a new hybrid learning mechanism that modifies the core optimization behavior of the agent changing how knowledge is propagated across goals and action dimensions rather than serving as a heuristic performance enhancement. This mechanism allows the agent to self-stabilize learning in sparse-reward environments and maintain policy safety under uncertainty, which has not been previously achieved by standard HER-augmented or clipped TD3 variants.

III. PROBLEM FORMULATION

Formulating spacecraft autonomy requires first establishing a precise representation of the spacecraft's attitude, since it serves as the fundamental state variable for decision-making, control adaptation, and resilience under uncertainty [57]. The spacecraft's attitude describes the orientation of its body-fixed frame relative to the orbital reference frame, directly influencing navigation accuracy, control precision, and overall mission stability.

Several mathematical representations have been proposed for attitude description, including Euler angles, rotation

TABLE 5. Comparison of RL methods.

Method	Advantages	Disadvantages	Ref.
SAC	Adaptable to disturbances	High computation, needs stacked observations	[46]
PPO	Stable learning, reliable control	Computationally demanding	[54]
TD3	Stable, low energy use	Hyperparameter-sensitive	[53]
DDPG	Learns complex policies efficiently	Prone to instability	[53]
Q-learning	Model-free, adaptable to unknown dynamics	Computationally intensive, sparse rewards	[48]
A2C	Real-time adaptability, fault-tolerant	High computational resources, scalability issues	[49]
PPO stacked	Best overall performance	Adapts dynamically to mass variations	[50]
SAC stacked	More generalized approach	Adapts dynamically to mass variations	[50]
EGE	Guided exploration, fault-tolerant	Reliance on expert knowledge, computational overhead	[51]
TD3-PER	High sample efficiency, stable control	Reward function tuning challenges	[54]
PPO	High accuracy, fuel-efficient trajectories	High sample demand, hyperparameter sensitivity	[52]

matrices, quaternions, Rodrigues parameters (RP), and modified Rodrigues parameters (MRP). Euler angles provide intuitive angular descriptions but suffer from singularities, while rotation matrices enable direct transformations but are computationally redundant. Quaternions avoid singularities and offer a compact formulation but can introduce sign ambiguities, whereas RP provides a vector-based approach with limited rotational coverage. MRPs, by contrast, can represent eigenaxis rotations up to 360 degrees with minimal singularities, combining efficiency with robustness [58], [59].

In this study, MRPs are adopted as the preferred attitude representation. Their smooth parameterization supports reinforcement learning by providing stable state features, facilitating generalization across diverse mission conditions, and enhancing policy resilience under dynamic and uncertain environments [60].

A. SPACECRAFT ATTITUDE KINEMATICS AND DYNAMICS

In the context of spacecraft kinematics and dynamics, satellite kinematics is the study of how a satellite's orientation changes over time, focusing on its rotation without considering forces [61]. The kinematic equation in terms of MRPs is given by [62]:

$$\dot{\rho} = G_{\rho}\omega \quad (1)$$

where $\omega \in \mathbb{R}^{3 \times 1}$ is the angular velocity vector relative to the inertial frame, and G_{ρ} represents a kinematic transformation matrix that relates the time derivative of the MRPs, $\dot{\rho}$, to the spacecraft's angular velocity vector ω and calculated as follows:

$$G_{\rho} = \frac{1}{2} \left(I - S(\rho) + \rho\rho^T - \frac{1}{2}\rho^T\rho I \right), \quad (2)$$

where I is the identity matrix of appropriate dimensions, $S(\cdot)$ represents a skew-symmetric matrix. The MRPs are defined

as $\rho_i = q_i/(1+q_0)$, where $\bar{q} = [q_0, q^T]^T \in \mathbb{R} \times \mathbb{R}^3$ represents the spacecraft quaternions, and $q = [q_1, q_2, q_3]^T$. Thus, MRPs are $\rho = [\rho_1, \rho_2, \rho_3]^T$.

While satellite dynamics focuses on a satellite's motion, including orientation changes (kinematics) and the effects of forces to ensure stability and control in space [61], the rotational motion of a rigid spacecraft with fully functioning actuators is governed by the following dynamics equation [62], [63]:

$$J\dot{\omega}_t - S(\omega_t)J\omega_t = u_t \quad (3)$$

In this equation, $J \in \mathbb{R}^{3 \times 3}$ is the spacecraft's moment of inertia matrix, which is symmetric and positive-definite, reflecting how the mass is distributed about its three principal axes. The term $\omega_t \in \mathbb{R}^3$ represents the spacecraft's angular velocity vector at time t , and $\dot{\omega}_t$ is its time derivative—i.e., the angular acceleration.

The term $S(\omega_t)J\omega_t$ accounts for gyroscopic torques (Coriolis effects), where $S(\omega_t)$ is the skew-symmetric matrix that performs the cross product operation with ω_t . This captures the nonlinear coupling between rotational axes during angular motion.

The right-hand side, $u_t = [u_1, u_2, u_3, u_4] \in \mathbb{R}^4$, represents the total control torque generated by four reaction wheels (RWs). Each RW contributes torque about a specific axis by changing its spin rate. The wheels operate within a rotational speed range of $[-1500, 1500]$ revolutions per minute and have a moment of inertia of $4.67 \times 10^{-4} \text{ kg}\cdot\text{dm}^2$, enabling fine-grained control of the spacecraft's angular momentum. The control torque u_t is implemented via high-bandwidth Reaction Wheel Assemblies (RWAs). RWAs are capable of executing high-frequency torque commands to effectively manage micro-vibrations and overcome nonlinearities such as zero-speed crossing friction, ensuring the high-precision pointing required for agile missions [64].

The control system aims to align the spacecraft's current orientation represented using MRP with a desired target orientation. The difference between the current and target orientation is denoted as the attitude error $\text{MRP}_{\text{error}}$, which is favored for its compactness and ability to avoid singularities.

To formally model and solve the spacecraft attitude control problem, a Markov Decision Process (MDP) formulation is introduced in the next section.

B. MDP FORMULATION

Spacecraft attitude control and fault recovery is framed as MDP. This formulation enables the application of DRL to develop optimal strategies for maintaining the spacecraft's orientation. Within this context, the spacecraft is modeled as an intelligent agent that interacts with its environment, adapting its actions to ensure stability even under challenging conditions, such as an RW failure.

The MDP is defined as:

$$\text{MDP} = (S, A, T, P, r) \quad (4)$$

A MDP is defined as a tuple (S, A, T, P, r) , where: The state space S represents all potential configurations of the system. The action space A encompasses all possible decisions the agent can make. The set T , which is a subset of $\mathbb{N}$, represents a sequence of time steps during which the agent interacts with the environment. The transition function $P(s'|s, a)$ defines the probability of transitioning to a state s' given the current state s and action a . The reward function $r(s, a)$ specifies the immediate reward obtained by the agent when it takes action a in state s .

The **state space** at time t , denoted as s_t , includes key parameters representing the spacecraft's attitude error and angular velocity:

$$s_t = \{\text{MRP}_{\text{error}}, \omega\} \quad (5)$$

where:

- $\text{MRP}_{\text{error}}$ represents the MRP that quantify the orientation error between the current and desired orientations. This choice enables the state vector to describe the attitude error relative to the target orientation in a simple, compact form, avoiding the complexity and potential singularities associated with other representations, such as Euler angles. By using MRPs, we can maintain a consistent, relative frame for error calculation, streamlining real-world implementation by removing any reliance on an arbitrary inertial reference frame [65].
- ω is the angular velocity of the spacecraft relative to an inertial frame, a parameter crucial for tracking the spacecraft's rotational dynamics. Specifically, ω provides information on the instantaneous rotation rate of the spacecraft's body with respect to a fixed external reference. This velocity measurement allows the control system to make corrections that stabilize and adjust the spacecraft's attitude as it seeks the target orientation. Including ω in the state vector is essential

for capturing the spacecraft's current motion state, enabling the agent to understand both the direction and magnitude of rotations. This, in turn, supports effective control strategies for damping oscillations and precisely orienting the spacecraft, particularly under conditions where reaction wheel limitations may impact stability

The **action space** a_t includes the commands applied to the RWs:

$$a_t = \{\tau_1, \tau_2, \tau_3, \tau_4\} \quad (6)$$

where τ_i is the torque applied to the RW. for example in case of a fault where a RW becomes unresponsive, the control and recovery system must redistribute torque across the remaining functional wheels and activate a backup RW to maintain control and stability. The goal of the control policy $\pi(s_t)$ is to minimize attitude error and regulate angular velocity. To achieve this, we employ the TD3 with HER and DWC, which stabilizes learning by clipping control inputs, especially in fault conditions that could destabilize the spacecraft.

To train the DRL agent and simulate the spacecraft's environment, we use the *Basilisk* simulation framework. Basilisk provides a high-fidelity environment to simulate spacecraft dynamics and control subsystems, allowing for robust training of the DRL agent under dynamic and fault-prone conditions [66]. By simulating real-time spacecraft operations, including RW faults, as illustrated in Figure 2, Basilisk enables the agent to interact with the spacecraft's control system, adapt its actions, and maintain stability. Using Basilisk as the simulation environment, this approach allows the spacecraft to autonomously manage faults, adapt control and recovery actions, and maintain satellite attitude in unpredictable space environments.

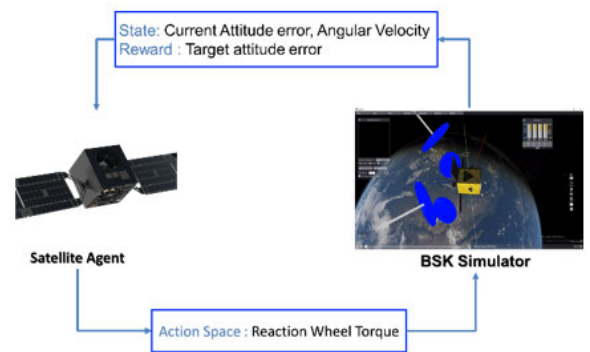


FIGURE 2. Interaction of agent with Environment via Basilisk.

IV. PROPOSED APPROACH

This study introduces the TD3-HD agent, a reinforcement learning framework designed to enhance spacecraft autonomy under uncertainty and dynamic operational conditions. Spacecraft often face challenges such as subsystem degradations, and structural reconfigurations, which demand control strategies that adapt in real time without human intervention.

The TD3-HD agent extends the TD3 algorithm by incorporating two key mechanisms. First, DWC constrains policy updates along individual action dimensions, mitigating instability and preventing unsafe actions during periods of high variability. Second, HER improves learning efficiency in sparse-reward environments by relabeling failed trajectories with achieved goals, enabling the agent to extract meaningful feedback even from unsuccessful episodes.

By integrating DWC and HER, TD3-HD strengthens stability, accelerates convergence, and improves robustness against uncertainties, providing a fault-tolerant and resilient control framework for autonomous spacecraft operations.

A. TD3

TD3 is an off-policy, actor-critic, policy gradient algorithm that extends (DDPG) [42], a proven method for continuous control applications. However, DDPG often experiences instability and exhibits high sensitivity to hyperparameter configurations, primarily due to value function overestimation in the critic network. This overestimation can accumulate during training, leading the agent to converge to suboptimal policies that may fail under uncertain conditions. TD3 addresses these fundamental challenges through three key innovations: (1) employing dual critic networks for value estimation, (2) implementing delayed policy updates, and (3) incorporating target policy smoothing through action noise regularization [12].

These enhancements significantly improve learning stability and robustness, making TD3 particularly suitable for spacecraft control applications where parameter uncertainties and environmental variations demand consistent performance across diverse operating conditions.

B. HER

HER effectively addresses the fundamental challenge of sparse reward signals in reinforcement learning by transforming suboptimal episodes into valuable learning experiences [67]. In spacecraft attitude control tasks under uncertain conditions particularly when dealing with actuator performance variations or system parameter changes the agent may receive minimal feedback if rewards are only provided upon achieving precise stabilization. This sparse feedback severely limits the agent's ability to develop robust control strategies during early training phases.

HER overcomes this limitation by subsequently redefining trajectory goals to match outcomes the agent actually achieved, such as intermediate attitude states or partial stabilization conditions. This approach converts apparent failures into simulated successes, significantly expanding the effective reward space and accelerating convergence while improving the agent's adaptability to uncertain operational scenarios.

Compared to alternative sparse reward mitigation techniques, HER demonstrates superior effectiveness for uncertainty management applications. Intrinsic motivation

methods, such as curiosity-driven exploration, can promote broader state-space exploration but often lack task-specific focus required for precision control under uncertain conditions [68]. Curriculum learning approaches, where agents progress through increasingly complex scenarios, show promise but require extensive manual tuning and domain expertise to define appropriate training progressions [69]. In contrast, HER provides a robust, plug-and-play solution that dynamically leverages the agent's experience to improve learning efficiency without requiring manual reward engineering or additional exploratory mechanisms. This makes HER particularly well-suited for space applications where stability and robustness under uncertain conditions are paramount.

C. DWC

To maintain both stability and safety during parameter updates in uncertain environments, traditional approaches restrict updates within predefined ranges to prevent excessively large changes that could destabilize the system. However, uniform clipping restrictions may degrade gradient information, limiting the effectiveness of policy learning [70]. This challenge is amplified when controlling multiple actuators or channels under uncertainty, where uniform bounds cannot capture the heterogeneous safety margins required across dimensions [71].

DWC addresses this limitation by applying adaptive clipping individually across action dimensions, ensuring that updates remain within safety-critical limits while preserving informative gradients. In this way, DWC improves policy stability, enhances convergence efficiency, and enforces safe behavior under dynamic and uncertain operational conditions.

To overcome these fundamental limitations, TD3-HD integrates DWC, an advanced technique introduced by [72]. DWC applies independent gradient clipping to each dimension of the action space, enabling selective constraint enforcement while preserving learning signals in stable dimensions.

In the DWC implementation:

- **Parameters clipped:** DWC applies independent clipping to each dimension of the policy gradient $\nabla_{\theta} J(\theta)$, where each dimension corresponds to individual actuator control adjustments.
- **Clipping mechanism:** For each dimension i , the gradient component $\nabla_{\theta} J(\theta)_i$ is clipped to a range $[-c_i, c_i]$, where c_i represents the dimension-specific clipping threshold. This approach ensures each component is constrained within appropriate bounds while maintaining learning capability across the entire action space.
- **Threshold determination:** The clipping thresholds c_i are established based on:
 - Adaptive thresholds calibrated to historical gradient variance in each dimension
 - Physical constraints of each actuator, including operational limits and safety margins

- Expected parameter variation ranges for different uncertainty scenarios, including system reconfigurations

DWC independently manages parameter adjustments for each control channel, preventing destabilizing updates without introducing gradient vanishing issues in unaffected dimensions. This preservation of learning signals in stable dimensions enables continued training efficiency even when some actuators require significant adaptation while others need minimal corrections. This methodology significantly enhances TD3-HD's sample efficiency by maintaining both stability and responsiveness under uncertain operating conditions, making it highly effective for robust spacecraft attitude control applications.

D. PRECISION CONTROL, MINIMAL EFFORT, AND CHATTERING SUPPRESSION IN TD3-HD

The ability of TD3-HD to maintain high control precision while minimizing control effort and avoiding chattering arises from the complementary interaction between TD3's policy smoothing, DWC's dimension-wise constraint enforcement, and HER's stabilization of the learning signal. Unlike classical controllers or unconstrained learning-based policies, TD3-HD explicitly regulates both *how fast* and *how much* each control channel is allowed to change during learning and execution. First, control precision is preserved through TD3's delayed policy updates and target policy smoothing [12]. Delayed actor updates prevent the policy from overreacting to transient critic estimation errors, while target policy smoothing suppresses high-frequency action variations by injecting bounded noise during critic training [12]. This combination ensures that learned control actions vary smoothly across time, enabling accurate attitude tracking without aggressive corrective torques.

Second, minimal control effort is achieved through DWC, which directly constrains policy gradient updates along individual action dimensions [72]. By limiting the magnitude of updates for each actuator independently, DWC prevents excessive torque commands that would otherwise arise from overcompensation or cross-axis coupling effects. Importantly, this selective clipping allows stable actuators to retain fine-grained control authority while restricting only those channels experiencing high uncertainty or abrupt disturbances [72]. As a result, TD3-HD redistributes control effort efficiently rather than increasing overall actuation energy.

Third, chattering suppression emerges naturally from the combined effect of smooth policy updates and bounded action corrections. In contrast to aggressively tuned feedback controllers, TD3-HD avoids rapid sign changes or oscillatory torque commands because DWC limits abrupt gradient-induced policy shifts [72], while TD3's policy smoothing penalizes high-frequency action components during training [12]. This prevents the agent from learning oscillatory control strategies even during abrupt structural reconfigurations.

Finally, HER indirectly contributes to stable and non-chattering behavior by improving learning consistency in sparse-reward settings [67]. By relabeling failed trajectories as successful with respect to achieved intermediate states, HER reduces noisy gradient updates caused by sparse terminal rewards. This stabilizes the critic estimates early in training, preventing erratic policy updates that could otherwise translate into oscillatory or overly aggressive control actions.

Together, these mechanisms ensure that TD3-HD achieves precise attitude regulation with smooth torque profiles, minimal unnecessary actuation, and no observable chattering.

E. ANALYTICAL JUSTIFICATION OF TD3-HD ADAPTABILITY

The adaptability improvements achieved by TD3-HD arise from explicit modifications to the policy-gradient learning dynamics rather than empirical tuning. Let $Q^*(s, a)$ define the true action-value function and two learned critics be

$$\widehat{Q}_i(s, a) = Q^*(s, a) + \varepsilon_i(s, a), \quad i \in \{1, 2\},$$

where ε_i denotes the estimation error. Let the actor be π_θ and the (approximate) policy gradient as [12]

$$\nabla_\theta J(\theta) \approx \mathbb{E}_s[\nabla_a \widehat{Q}(s, a)|_{a=\pi_\theta(s)} \nabla_\theta \pi_\theta(s)].$$

Lemma 1 (TD3: Non-Positive Expected Target Bias [12]): Assume $\varepsilon_1(s, a)$ and $\varepsilon_2(s, a)$ are identically distributed with $\mathbb{E}[\varepsilon_i(s, a)] = 0$ and have finite expectation. Forming the TD3 Bellman target using minimum of the two critics yields an expected target bias that is non-positive:

$$\mathbb{E}[\min(\widehat{Q}_1, \widehat{Q}_2) - Q^*] = \mathbb{E}[\min(\varepsilon_1, \varepsilon_2)] \leq 0,$$

Proof 1: Because $\min(\varepsilon_1, \varepsilon_2) \leq \varepsilon_i$ for $i = 1, 2$ and $\mathbb{E}[\varepsilon_i] = 0$, it follows that $\mathbb{E}[\min(\varepsilon_1, \varepsilon_2)] \leq \mathbb{E}[\varepsilon_i] = 0$. Hence, on average, the TD3 min-operator prevents systematic positive overestimation of targets; this reduces the chance that actor updates exploit optimistic critic errors and thus produces policy updates that are more reliably aligned with increasing true return.

Lemma 2 (HER: Effective Increase in Informative Samples [13]): Consider a goal-conditioned task where, for a defined goal g , a sampled trajectory yields a binary reward $R \in \{0, 1\}$ with success probability $p \ll 1$. Without HER, the empirical variance of the reward signal is $p(1-p) \approx p$, and the proportion of informative (nonzero) gradient contributions equals p . With HER relabelling that converts a subset of previously failed trajectories into successful examples for other goals, the effective fraction of informative samples increases to $p_{\text{eff}} > p$. Consequently, the expected number of informative gradient contributions per unit of data increases by p_{eff}/p , improving the signal available to the gradient estimator and accelerating learning in sparse regimes.

Proof 2: The variance of a Bernoulli reward is $p(1-p) \approx p$ and the number of nonzero reward samples in a dataset of size N is N_p . HER increases N_p to $N_{p_{\text{eff}}}$ by relabelling, so the count of informative updates scales up by

p_{eff}/p . While the per-sample variance may change slightly, the increase in informative samples reduces the effective sampling noise in gradient estimation (more nonzero gradient contributions per batch), which yields faster convergence in practice.

Lemma 3 (Dimension-Wise Clipping: Variance Control in High Dimensions [72]): Let an importance or update weight for a D -dimensional action factor approximately as

$$w = \prod_{d=1}^D w_d.$$

Scalar clipping of the full product $w \leftarrow \min(w, c)$ can introduce large bias when a single w_d is extreme. If instead each per-dimension component is clipped $w_d \leftarrow \min(w_d, c_d)$ and the w_d are (approximately) independent or weakly correlated, then the variance of the clipped weight is bounded by a sum of per-dimension variances. This per-dimension bound grows more slowly with D than the variance of the product with scalar clipping, yielding improved stability when reusing off-policy/HER samples in high-dimensional action spaces.

Proof 3: For (approximately) independent components

$$\text{Var}\left(\prod_{d=1}^D \tilde{w}_d\right)$$

can be upper bounded by terms involving sums/products of per-dimension variances (use expansions via the product moment formulas). Clipping each $\tilde{w}_d$ individually reduces extreme tails per coordinate, controlling higher moments that dominate the product's variance. Heuristically, bounding each coordinate prevents a single dimension from driving an exponential blow-up in variance that scalar clipping of the product cannot prevent.

Combined Effect: 3 above Lemmas together describe complementary mechanisms: TD3 reduces optimistic critic bias (safer gradient directions), HER increases the density of informative updates (stronger learning signal in sparse regimes), and dimension-wise clipping bounds per-coordinate extremes (safer reuse of heterogeneous replay). The joint effect is reduced effective bias, reduced effective variance, and improved sample efficiency—properties that enhance adaptability to changing goals or dynamics.

F. DOMAIN-SPECIFIC DESIGN FOR SATELLITE FAULT RECOVERY

While TD3, HER and DWC provide strong off-policy learning primitives, satellite fault recovery imposes constraints and objectives that require specific algorithmic adaptations. We summarize the design principles implemented in TD3-HD and their motivation:

- **Goal-relabeling strategy:** goal relabeling adapts standard HER for fault recovery by restricting relabeled goals to meaningful, reachable, and safety-relevant outcomes (e.g., nominal attitude, safe pointing, or reaction-wheel

de-saturation states) rather than arbitrary visited states. The goal-relabeling strategy replaces each episode's original target attitude with the actually achieved attitude and angular-velocity states from future time steps; each relabeled transition is then re-evaluated with an updated reward, allowing the agent to reinterpret unsuccessful trajectories as successful relative to achievable intermediate goals. By constraining relabeling to these operationally useful recovery states, the agent is trained to seek safety-aware, mission-relevant recoveries instead of learning to reach arbitrary visitable states.

- **Per-actuator (dimension-wise) clipping normalized by hardware limits:** We clipped per-dimension importance/updates using each actuator's physical limits (torque/current). This prevents a single saturated actuator (e.g., a reaction wheel near saturation) from producing extreme updates that destabilize learning, which is a key concern in satellite control.
- **Energy- and mission-aware reward regularization:** Recovery actions are penalized relative to their power cost and mission impact (e.g., lost observation time), enabling the agent to choose recovery strategies that balance health versus mission objectives.

Above mentioned adaptation address general satellite issues — actuator saturation/failure, limited compute and power, safety-critical operation, and multi-objective tradeoffs — and show how TD3/HER/clipping are applied in designing DRL principles specific to satellite fault recovery.

G. CONTROL AUTHORITY REDISTRIBUTION

When system uncertainties impact actuator performance or availability, TD3-HD adaptively redistributes control authority to preserve attitude precision and system stability. The agent dynamically adjusts the relative contribution of each actuator, ensuring that control efforts reflect their current effectiveness and operational health.

The total control torque T applied to the spacecraft is given by:

$$T = \sum_{i=0}^3 \lambda_i R W_i \quad (7)$$

where $R W_i$ denotes the torque contribution from the reaction wheel, and λ_i represents the adaptive policy weight governing its share of control authority.

These weights λ_i serve as dynamic scaling factors that evolve according to real-time performance feedback. Under nominal conditions, they maintain a balanced distribution across actuators, assuming symmetric capabilities. When uncertainties such as partial degradation, saturation, or intermittent availability occur, the corresponding λ_i values are reduced, redistributing control demand across healthier actuators. This continuous adaptation mitigates instability and prevents overloading of vulnerable subsystems.

By dynamically reallocating authority the TD3-HD agent achieves resilient adaptation across a spectrum of

uncertainties. This ensures smooth transitions, minimizes performance loss, and maintains precise attitude control without manual intervention, even in degraded operational states.

H. REWARD FUNCTION DESIGN

The reward function serves as a fundamental guidance mechanism in reinforcement learning, directing agent behavior toward achieving specific control objectives under uncertain conditions [73]. It provides quantitative feedback by assigning numerical rewards to actions and states, reflecting how effectively the agent's behavior aligns with desired performance goals while managing system uncertainties.

Our reward function is specifically designed to minimize attitude error while ensuring stable control behavior under uncertain conditions, comprising three integrated components:

- **Attitude Error Reduction Reward:** The reward function for attitude error reduction is defined as:

$$r_1 = e_{\text{previous}} - e_{\text{current}} \quad (8)$$

where:

- e_{previous} represents the attitude error at the previous time step, reflecting the difference between desired and actual orientations
- e_{current} represents the current attitude error, enabling assessment of error reduction progress

This component encourages consistent error reduction while providing continuous feedback during uncertain operational phases.

- **Stability Preservation Penalty:**

$$r_2 = \begin{cases} -10, & \text{if } |\omega| > 1 \\ 0, & \text{otherwise} \end{cases} \quad (9)$$

where ω represents the angular velocity. Excessive angular velocities are penalized to maintain system stability and prevent oscillatory behavior under uncertain conditions.

- **Precision Maintenance Incentive:**

$$r_3 = \begin{cases} 0.01, & \text{if } e_{\text{current}} < 0.25 \\ -0.01, & \text{otherwise} \end{cases} \quad (10)$$

This component promotes precise attitude maintenance by rewarding low attitude errors. The threshold value of 0.25 degrees was selected based on precision spacecraft pointing requirements and validated through extensive testing [74]. This threshold represents an optimal balance between achievable performance under uncertain conditions and mission requirements for Earth observation and scientific applications.

The comprehensive reward function is:

$$\text{reward} = r_1 + r_2 + r_3 \quad (11)$$

This formulation ensures the spacecraft is rewarded for error reduction, penalized for unstable behavior, and incentivized

to maintain precise control under uncertain operational conditions.

I. ALGORITHM INTEGRATION AND IMPLEMENTATION

The TD3 algorithm, recognized for its stable learning characteristics and continuous control capabilities [75], is enhanced with DWC to independently manage parameter updates for each control channel. This mechanism effectively mitigates large and destabilizing updates during uncertain operational phases [72].

Additionally, HER addresses sparse reward challenges by reinterpreting suboptimal actions as successful experiences with modified objectives [76], enabling effective learning even under challenging uncertain conditions. The integrated approach is detailed in Algorithm 1, while the network architecture is illustrated in Figure 3.

The system integrates dual neural networks: an actor network and a critic network. The actor network, consisting of four specialized sub-networks parameterized by λ_i (for $i = 1, 2, 3, 4$), generates control commands for each actuator by outputting Gaussian parameters μ_i and σ_i , which are sampled to produce normalized control actions a_i . These actions are applied to the spacecraft's reaction wheels to maintain desired attitude under uncertain conditions.

During training, the spacecraft's state, represented as $[\text{MRP}_{\text{error}}, \omega]$ (MRP for attitude error and angular velocity vector), is stored in the HER buffer E , which transforms unsuccessful trajectories by replacing original goals with achieved outcomes. The TD3-HD method employs batch training, where the actor network updates its policy based on sampled transitions from E and maintains parameter stability through delayed updates.

DWC is applied to limit excessive updates for each control channel independently, while Importance Sampling (IS) weights ρ_t correct for policy distribution mismatches. The critic evaluates candidate control actions by computing advantage values A_1 , which quantify action quality under uncertainty. During TD3 optimization these advantages, together with the IS weights and the IS policy loss J_{IS} , steer the update of the actor's Gaussian parameters to achieve precise attitude control despite system uncertainties. In our implementation DWC is applied directly to the policy gradients during actor updates: each gradient component associated with a control dimension is clipped independently within its threshold range $[-c_i, c_i]$. Importantly, no additional clipping is imposed on the actor's output actions beyond the environment's physical actuator limits.

V. EXPERIMENTAL SETUP

The Basilisk Astrodynamics Simulation Framework [66] provides a high-fidelity simulation environment for modeling the kinematics and dynamics of a small satellite in LEO. Hence, this simulator is used as simulation environment in our experiments. The satellite's orientation is represented by MRP and controlled using four Honeywell HR16 RWs arranged in a pyramid configuration. These RWs deliver

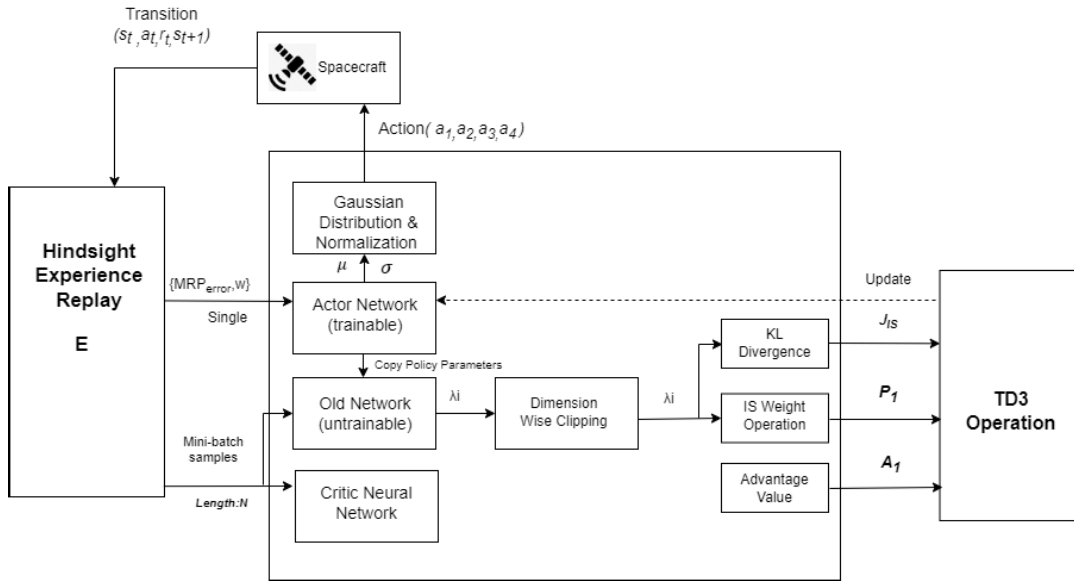


FIGURE 3. TD3-HD architecture.

torque adjustments to maintain the desired attitude of the satellite. To evaluate the fault tolerance of various control strategies, an RW failure is simulated by disabling one of the wheels at 3000th second. Key telemetry data, including MRP, angular velocity, and RW torque, are recorded over a 8000-second period.

This study evaluates advanced DRL algorithms for CubeSat attitude control, focusing on the proposed TD3-HD algorithm. TD3-HD enhances control performance under actuator faults by integrating HER for improved learning in sparse reward environments. Additionally, PPO [77], known for its robustness in complex tasks and stable policy updates, is assessed alongside A2C and TD3, which serves as a baseline for reliable CubeSat attitude control [54].

This comprehensive evaluation provides a comparative analysis of each algorithm's strengths and suitability for autonomous CubeSat attitude control.

All DRL algorithms in this study are implemented using the Stable-Baselines3 library [78], an open-source toolkit built on PyTorch that facilitates efficient and reliable model development for DRL research. For training the neural networks, a custom simulation environment is created in Python, adhering to the OpenAI Gym standards [79]. This standardization ensures a flexible interface for simulating interactions and enables seamless integration of the DRL algorithms. Figure 4 provides an overview of the simulation configuration.

Table 6 outlines the training parameters used for all the DRL algorithms in our benchmark. Both actor and critic networks comprise two hidden layers of 256 neurons each, with ReLU activations. The actor output uses tanh normalization to constrain torque commands, while the critic output is linear. The same architecture was used for PPO,

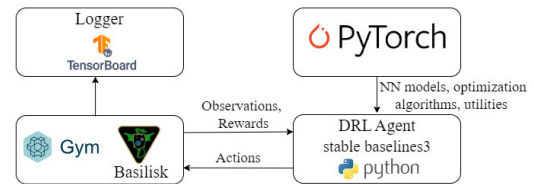


FIGURE 4. Simulation configuration used for satellite attitude control.

A2C, and TD3 to ensure consistent benchmarking; TD3-HD additionally includes four actor sub-networks corresponding to the reaction-wheel channels.

TABLE 6. Training parameters for DRL algorithms.

Parameter	Value
Learning Rate (ζ)	3×10^{-4}
Replay Buffer Size	1,000,000
Batch Size (M)	128
Target Update Interval	2
Clipping Thresholds (DWC)	0.2
Hidden Layers	2 (fully connected)
Hidden Units per Layer	256
Activation Functions	ReLU (hidden), tanh (actor output), linear (critic output)
IS (α_{IS})	1
Trajectory Size (N)	100
Actor Sub-networks (λ_i)	4

Figure 5 compares the training reward progression of A2C, PPO, TD3, and TD3-HD over the training episodes. All methods exhibit pronounced reward fluctuations during the early episodes of training due to exploration in a high-dimensional and constrained control environment. However, clear differences emerge in terms of convergence speed, reward stability, and sample efficiency.

Algorithm 1 TD3-HD for Uncertainty Management

- 1: **Input:** Maximum iterations L , epochs K , HER goal sampling strategy, DWC bounds for uncertainty management.
- 2: **Initialization:**
- 3: Initialize actor network weights λ_i for $i = 1, 2, 3, 4$ (corresponding to each control channel).
- 4: Initialize Q-value critic networks for TD3.
- 5: Initialize (IS) weighting factors $\alpha_{IS} = 1$, learning rate ζ , and DWC thresholds for uncertainty adaptation.
- 6: Initialize replay buffer E for storing state transitions $\{(\text{MRP}_{\text{error}}, \omega), a, r, (\text{MRP}'_{\text{error}}, \omega')\}$.
- 7: Load spacecraft dynamics model with uncertainty parameters and system variations.
- 8: **Main Loop: Uncertainty Adaptation Phase**
- 9: **for** iteration = 1 to L **do**
- 10: Initialize spacecraft states $[\text{MRP}_{\text{error}}, \omega]^T$ under uncertain conditions.
- 11: Load desired target states (goal orientation and angular velocity).
- 12: Store current actor network parameters $\lambda_i \leftarrow \lambda$ for stability reference.
- 13: Apply actor policy π_{λ_i} to sample uncertainty-adapted actions $a = [a_1, a_2, a_3, a_4]$.
- 14: Store state transitions (s_t, a_t, r_t, s_{t+1}) in replay buffer E , applying HER strategy to transform uncertain episodes into learning experiences.
- 15: Compute advantage estimates $\hat{A}_t$ using transitions from HER buffer E under uncertainty conditions.
- 16: **end for**
- 17: **Training Loop for Uncertainty Robustness**
- 18: **for** epoch = 1 to K **do**
- 19: **for** each gradient step **do**
- 20: Sample mini-batch of size M from replay buffer E .
- 21: Apply DWC to parameter updates λ_i for each control channel, managing uncertainty-induced variations.
- 22: Update actor network weights for uncertainty adaptation:

$$\lambda \leftarrow \lambda + \zeta \nabla_{\lambda} J_{IS}(\lambda)$$
 where J_{IS} represents IS-weighted policy loss optimized for uncertain conditions.
- 23: **end for**
- 24: Update IS weighting factor α_{IS} adaptively to balance experience replay under uncertainty.
- 25: **end for**

TD3-HD demonstrates the fastest convergence among all evaluated methods, reaching a stable and consistently high cumulative reward after approximately 65 training episodes. In contrast, standard TD3 and PPO require approximately 85–95 training episodes to reach comparable stability, while A2C does not fully converge within the 140 training episodes

shown. This indicates that TD3-HD converges with roughly 30–60% fewer training episodes than TD3 and PPO.

Beyond faster convergence, TD3-HD also exhibits substantially lower reward variance after convergence. Once stabilized, the TD3-HD reward curve remains smooth and consistent, whereas PPO converges to a lower asymptotic reward and A2C continues to show large oscillations. Standard TD3 achieves a similar final reward level to TD3-HD but with slower convergence and higher transient variance.

Overall, the results in Fig. 5 demonstrate that TD3-HD not only achieves the highest cumulative reward but does so with significantly fewer training episodes compared to PPO and standard TD3. This confirms that TD3-HD offers superior sample efficiency and faster convergence, making it particularly suitable for autonomous spacecraft control tasks where training efficiency is a critical constraint.

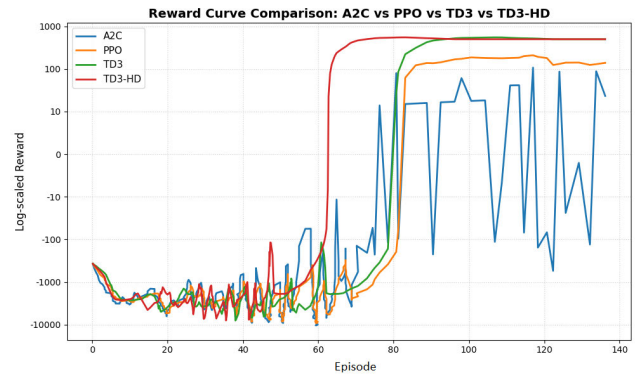


FIGURE 5. Comparison of selected training algorithms based on Reward Curves.

VI. EXPERIMENTAL RESULTS

This section presents a comprehensive experimental evaluation and comparative analysis of DRL algorithms for spacecraft attitude control and fault recovery under uncertain and dynamically varying operational conditions. The control and recovery system performance assessment was conducted using multiple quantitative metrics, with primary evaluation based on pointing accuracy measurements between the spacecraft's body-fixed coordinate system and inertial reference frames, quantifying the precision of three-axis orientation alignment under both nominal and perturbed conditions.

The robustness evaluation framework incorporated two distinct categories of system uncertainties representative of real-world operational challenges: actuator performance variations including degradation in reaction wheel characteristics and intermittent operational irregularities; structural configuration changes such as deployable component activation and mass distribution alterations that fundamentally modify system dynamics. The comprehensive experimental investigation demonstrates that the proposed TD3-HD algorithm exhibits exceptional performance characteristics across all evaluated uncertainty categories, demonstrating superior

adaptability through its integrated HER mechanism and precise actuator utilization via DWC, effectively managing uncertain conditions while maintaining control accuracy and operational stability superior to conventional controllers and alternative DRL approaches.

A. PERFORMANCE UNDER UNRESPONSIVE RW: SCENARIO 1

This section provides a comprehensive comparative analysis of PD controller and three DRL algorithms A2C, PPO, and TD3 in addressing attitude control under uncertain actuator performance conditions.

1) PD CONTROLLER: BASELINE PERFORMANCE ANALYSIS

The conventional PD controller serves as a baseline for evaluating the adaptive capabilities of advanced control algorithms under system uncertainties. Under nominal operating conditions, the PD controller demonstrates satisfactory attitude tracking performance, as evidenced by the alignment between desired and actual attitude trajectories. However, when subjected to system perturbations such as actuator performance degradation occurring at $t = 3000$ seconds the controller exhibits significant limitations.

Following the introduction of system uncertainty, the PD controller fails to maintain adequate performance without external intervention. The error history and angle error evolution (Figure 6) reveal substantial deviations from target attitudes, with persistent angular velocity oscillations indicating insufficient robustness to parameter variations. This behavior necessitates manual parameter readjustment to restore system stability, highlighting the fundamental limitation of fixed-gain controllers in uncertain environments.

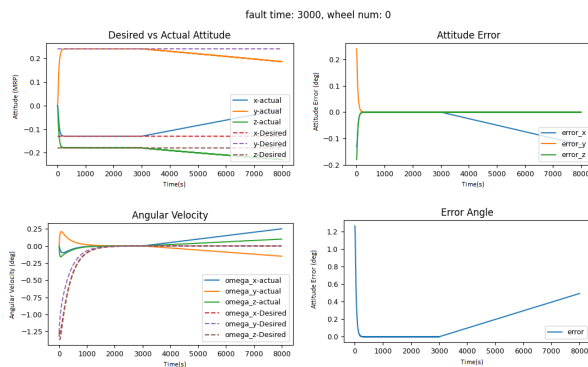


FIGURE 6. PD controller performance under system uncertainty introduced at $t = 3000$ seconds, showing attitude error evolution and angular velocity response degradation.

The torque distribution analysis (Figure 7) further illustrates the controller's inability to adapt to changing system conditions. When actuator performance is compromised, the remaining actuators cannot compensate effectively without manual reconfiguration or ground-based parameter

updates. This limitation emphasizes the critical need for autonomous adaptation mechanisms in modern spacecraft control systems.

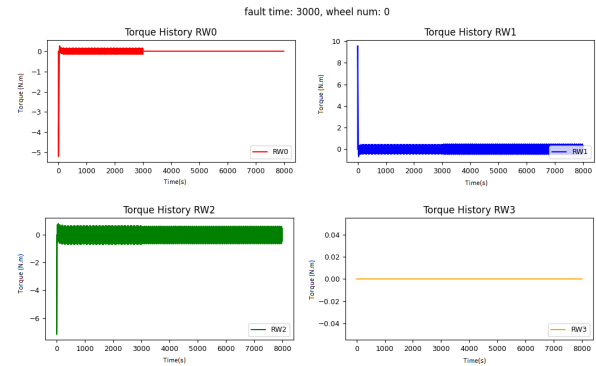


FIGURE 7. Reaction wheel torque distribution under PD control during system uncertainty, demonstrating inadequate compensation mechanisms.

These observations underscore the necessity for advanced, adaptive control mechanisms capable of autonomous response to system uncertainties. Unlike traditional fixed-parameter controllers, reinforcement learning algorithms possess the inherent capability to learn optimal control policies through experience, enabling real-time adaptation to changing system conditions without requiring external intervention.

2) PPO PERFORMANCE ANALYSIS

The PPO algorithm demonstrates reasonable adaptation capabilities when subjected to system uncertainty at $t = 3000$ seconds. As illustrated in Figure 8, the algorithm successfully maintains attitude alignment with desired trajectories despite degraded system conditions. The attitude error evolution indicates that PPO preserves acceptable precision levels across all rotational axes following the uncertainty introduction, demonstrating inherent robustness.

The angular velocity response reveals PPO's capability to manage dynamic variations, achieving convergence between actual and desired velocity profiles. However, the algorithm exhibits extended settling periods and occasional control variations due to its exploration mechanisms under rapidly changing system conditions, which may compromise performance in time-critical applications.

Regarding actuator utilization, PPO demonstrates adaptive torque redistribution capabilities among available reaction wheels when system conditions change (Figure 9). The algorithm autonomously adjusts torque allocation to compensate for degraded actuator performance, exhibiting increased activity fluctuations as it learns to manage the altered system dynamics. This adaptive redistribution highlights PPO's resilience in maintaining control authority under uncertain conditions.

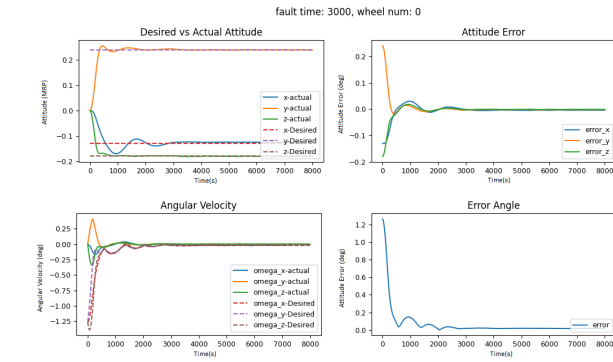


FIGURE 8. PPO algorithm performance under system uncertainty, showing attitude tracking and error evolution characteristics.

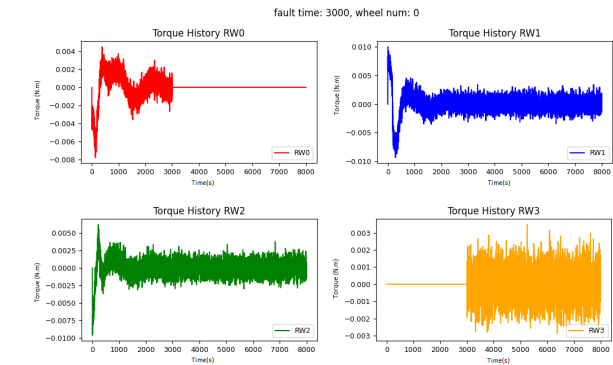


FIGURE 9. PPO torque redistribution strategy under system uncertainty, demonstrating adaptive actuator utilization.

3) A2C PERFORMANCE ANALYSIS

The A2C algorithm exhibits comparable performance to PPO under uncertain system conditions. As demonstrated in Figure 10, A2C maintains reasonable attitude tracking capabilities following the uncertainty introduction at $t = 3000$ seconds. The algorithm gradually minimizes deviations between actual and desired attitudes, showing satisfactory adaptability to changing system parameters.

The attitude error characteristics reveal rapid initial error reduction, particularly along critical axes, with stable low-error maintenance in the post-uncertainty phase. A2C effectively manages angular velocity stabilization, though it requires extended convergence periods when adapting to highly dynamic system variations. This behavior suggests limitations in rapid response scenarios where immediate adaptation is crucial.

The torque management analysis (Figure 11) reveals A2C's effective redistribution capabilities among available actuators to compensate for performance variations. The algorithm demonstrates reasonable adaptation to changing actuator characteristics, though with noticeable fluctuations during the transition period.

4) TD3 PERFORMANCE ANALYSIS

The standard TD3 algorithm demonstrates superior performance characteristics compared to policy gradient methods

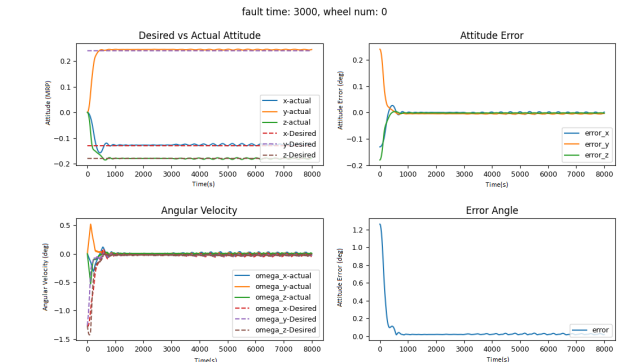


FIGURE 10. A2C algorithm response to system uncertainty, illustrating attitude control performance and error convergence patterns.

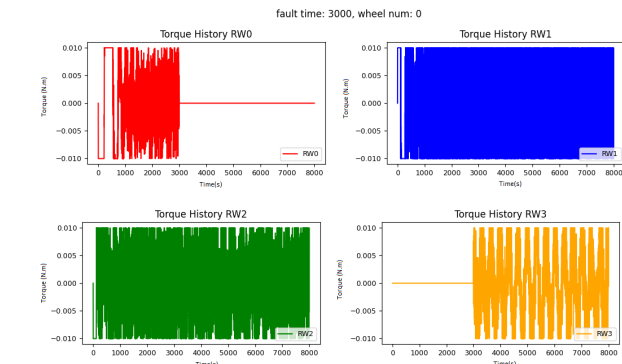


FIGURE 11. A2C torque adaptation strategy under system uncertainty, showing actuator utilization patterns.

when managing system uncertainties. As illustrated in Figure 12, TD3 exhibits rapid attitude alignment with desired trajectories, maintaining minimal deviations even during uncertain conditions introduced at $t = 3000$ seconds.

The attitude error evolution shows TD3's ability to achieve rapid error reduction across all rotational axes, sustaining near-zero error levels following system perturbations. The angular velocity response demonstrates exceptional stability with minimal oscillatory behavior, reflecting TD3's precision in managing dynamic uncertainties under challenging conditions.

TD3 achieves smooth torque redistribution among available actuators following system uncertainty introduction (Figure 13). This adaptive torque management demonstrates TD3's robustness in maintaining control authority and system stability during uncertain operating conditions. However, the gradual convergence characteristics observed reflect the challenges associated with sparse reward environments, requiring extended learning periods to achieve optimal performance.

5) COMPARATIVE PERFORMANCE SUMMARY: SCENARIO 1

All evaluated algorithms PPO, A2C, and TD3 demonstrate reasonable adaptability to system uncertainties, though

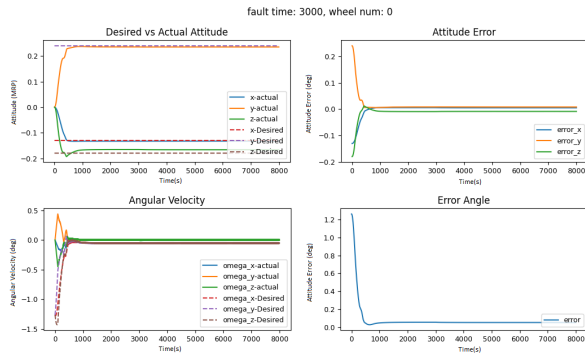


FIGURE 12. TD3 algorithm performance under system uncertainty, demonstrating superior attitude control precision and stability.



FIGURE 13. TD3 torque redistribution under system uncertainty, showing balanced actuator utilization and adaptation capabilities.

each exhibits distinct characteristics and limitations. PPO provides moderate adaptation capabilities but may introduce oscillatory behaviors under persistent uncertain conditions. A2C offers resilience and flexible torque redistribution, though with slower convergence rates and higher transient fluctuations. TD3 excels in precision and stability for both attitude control and torque management, though it requires higher computational resources and may exhibit sensitivity to control over-adjustments.

Overall, TD3 presents the most robust solution among traditional DRL approaches for uncertain system conditions. However, the gradual convergence characteristics observed across all algorithms highlight the challenges posed by sparse reward environments, necessitating extended learning periods to achieve stable control performance. These comparative results underscore the need for enhanced algorithmic approaches that can overcome these fundamental limitations.

6) PERFORMANCE OF THE PROPOSED TD3-HD ALGORITHM

Figures 14 and 15 present the performance characteristics of the proposed TD3-HD algorithm, incorporating HER and DWC enhancements. This advanced TD3 variant addresses

the fundamental limitations of standard reinforcement learning approaches while demonstrating superior performance under system uncertainties.

The integration of HER enables TD3-HD to learn effectively from sparse reward environments, while DWC provides refined control precision through dimensional torque adjustment limitations. These enhancements significantly improve resilience and adaptability under challenging conditions, including the system uncertainty introduced at $t = 3000$ seconds.

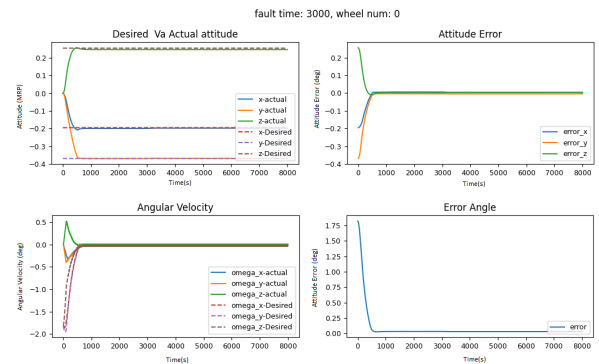


FIGURE 14. TD3-HD algorithm performance under system uncertainty, demonstrating enhanced attitude control precision and rapid adaptation capabilities.

The attitude tracking analysis reveals TD3-HD's rapid alignment with target orientations, maintaining exceptional stability and precision throughout the simulation period. The attitude error evolution demonstrates consistent minimization across all rotational axes, achieving persistently low error levels following uncertainty introduction, reflecting enhanced control accuracy compared to baseline algorithms.

The angular velocity response illustrates TD3-HD's superior damping characteristics, achieving near-instantaneous alignment between actual and desired velocity profiles. The error angle convergence analysis further emphasizes the control precision of TD3-HD, with attitude errors rapidly converging to stable, near-zero values, underscoring the algorithm's robustness under uncertain system conditions.

The torque management analysis (Figure 15) highlights TD3-HD's efficient actuator utilization strategies. The algorithm demonstrates smooth and dynamic torque redistribution among available actuators when system conditions change. The DWC mechanism ensures dimensionally constrained torque outputs, preventing overcompensation while enabling balanced control authority utilization.

This approach provides significantly smoother control characteristics compared to PPO, A2C, and standard TD3, as evidenced by the consistent and stable torque responses across all actuators. The superior performance characteristics establish TD3-HD as an exceptional solution for autonomous attitude control under uncertain operating conditions.

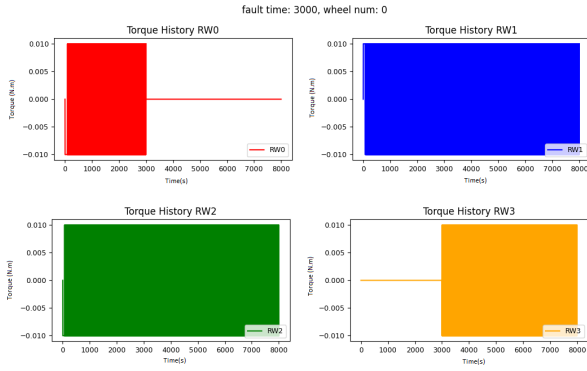


FIGURE 15. TD3-HD torque management under system uncertainty, showing optimal actuator utilization and balanced control distribution.

B. STRUCTURAL RECONFIGURATION ANALYSIS: SCENARIO 2

To evaluate adaptability under dynamic system variations, we simulate a structural reconfiguration that abruptly changes the spacecraft inertia matrix at $t = 3000$ s, mimicking solar panel deployment, antenna extension, or payload reconfiguration. Such changes alter system dynamics and increase sensitivity to control torques and disturbances [77]. The objective is to assess each controller's ability to maintain stability and precision without retraining or manual gain updates.

1) PD CONTROLLER: BASELINE PERFORMANCE ANALYSIS

As a classical baseline, the Proportional-Derivative (PD) controller struggles to accommodate the abrupt inertia change (Figs. 16–17). Immediately after $t = 3000$ s, pronounced oscillations and persistent attitude error are observed, with delayed damping and overshoot. Performance indicates that acceptable tracking would require *retuning* the gains for the new dynamics, underscoring the vulnerability of fixed-gain linear control under nonlinear uncertainties and parameter shifts [80].

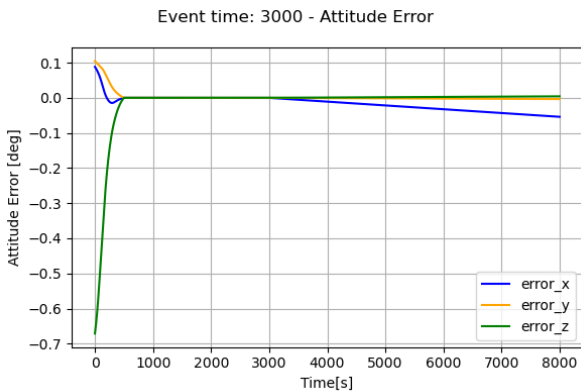


FIGURE 16. PD response to structural reconfiguration: attitude tracking and adaptation characteristics.

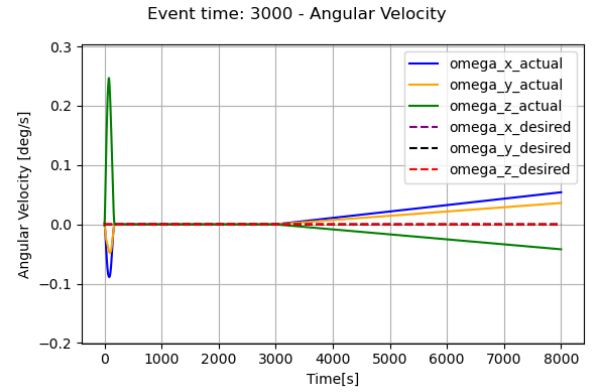


FIGURE 17. PD detailed metrics during reconfiguration: angular velocity and attitude error evolution.

2) PPO PERFORMANCE ANALYSIS

PPO handled the fault effectively but shows moderate overshoot (Figs. 18–19). Critically, the peak attitude error rises to ≈ 0.30 at the reconfiguration time, exceeding the mission's acceptable limit of 0.25 [74]. This indicates insufficient robustness to sudden parameter changes despite eventual recovery.

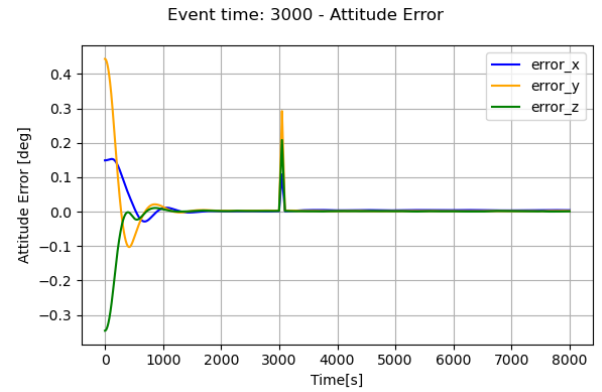


FIGURE 18. PPO response to structural reconfiguration: attitude control and adaptation.

3) A2C PERFORMANCE ANALYSIS

A2C successfully reorients to the target attitude (Figs. 20–21), but exhibits noticeable fluctuation. The persistent ripple can reduce actuator effectiveness over long horizons and degrade imaging resolution, making A2C less suitable when tight pointing stability is required immediately after configuration changes.

4) TD3 PERFORMANCE ANALYSIS

TD3 achieves reasonable stabilization following the event (Figs. 22–23). However, it requires more time to settle at the beginning, attributable to exploration and conservative target policy updates. Indicating room for improvement in handling abrupt structural changes [53].

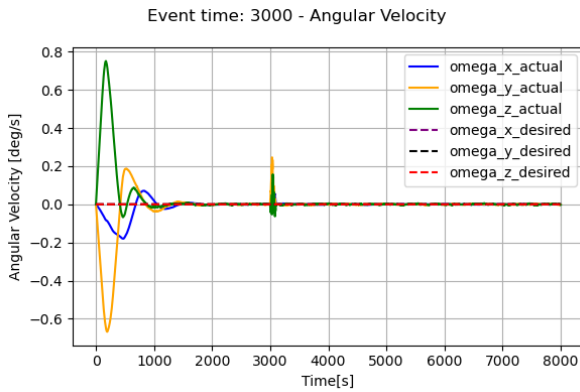


FIGURE 19. PPO detailed metrics during reconfiguration: angular velocity and attitude error.

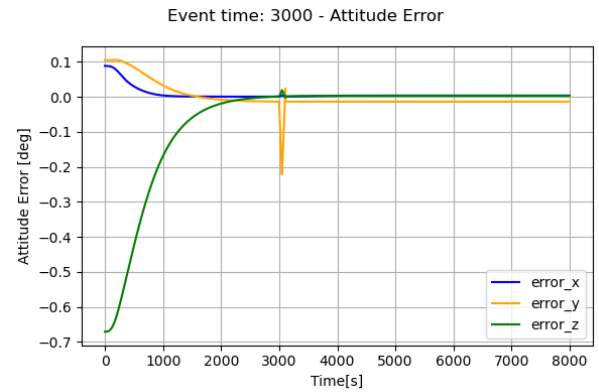


FIGURE 22. TD3 performance during reconfiguration: attitude tracking and adaptation.

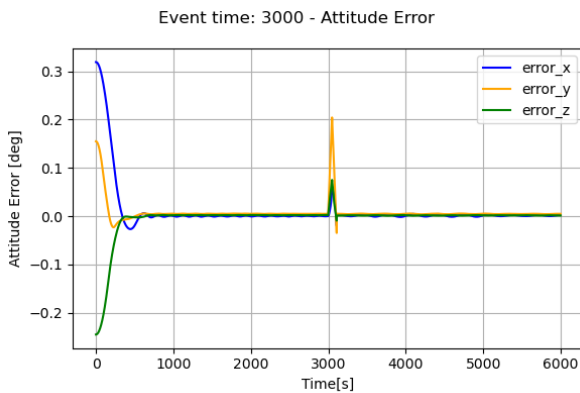


FIGURE 20. A2C performance during reconfiguration: attitude tracking and adaptation.

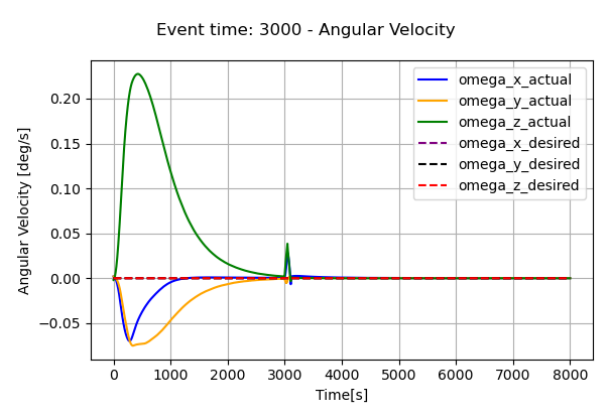


FIGURE 23. TD3 detailed metrics during reconfiguration: system response characteristics.

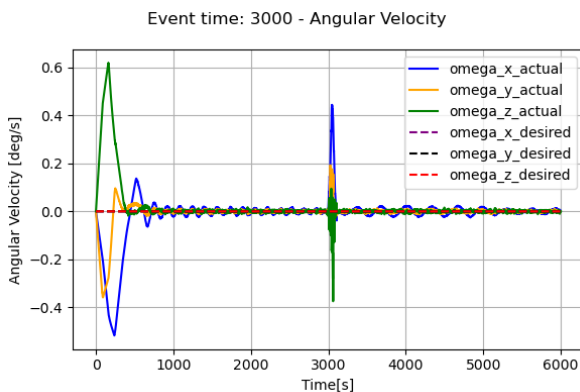


FIGURE 21. A2C detailed metrics during reconfiguration: system response characteristics.

5) COMPARATIVE PERFORMANCE SUMMARY: SCENARIO 2

The comparative evaluation highlights clear distinctions between classical and learning-based controllers under structural reconfiguration. The Proportional-Derivative (PD) controller demonstrates significant limitations, with pronounced oscillations, overshoot, and persistent attitude error,

requiring manual retuning to remain viable. This underscores the fragility of fixed-gain approaches under nonlinear uncertainties and parameter shifts.

Among reinforcement learning methods, PPO exhibits effective recovery but suffers from a critical overshoot, with peak attitude error exceeding mission tolerance. This reveals that although PPO can redistribute control effectively, its robustness to abrupt parameter changes is insufficient for stringent pointing requirements. A2C achieves successful reorientation but introduces persistent oscillations that may impair long-term actuator efficiency and degrade payload performance, limiting its applicability to precision-oriented missions.

TD3 achieves stable tracking following reconfiguration and demonstrates improved robustness compared to PPO and A2C. However, its conservative policy update mechanism results in slower initial convergence, leaving room for improvement in rapidly adapting to abrupt structural variations.

Overall, the results indicate that while deep reinforcement learning methods surpass the PD baseline in adaptability, differences in transient response, overshoot, and stability

margins highlight the trade-offs between policy robustness and adaptability across algorithms.

6) PERFORMANCE OF THE PROPOSED TD3-HD ALGORITHM

The proposed TD3-HD delivers the best overall performance (Figs. 24–25). It maintains peak attitude error within the 0.25 limit at the reconfiguration time, exhibits the fastest convergence to near-zero error, and shows the smoothest angular-rate decay with minimal oscillation across axes. Torque histories indicate balanced redistribution without excessive fluctuations, promoting energy-efficient control throughout the transition. Collectively, these results demonstrate superior autonomous adaptation to structural changes compared with PD, PPO, A2C, and vanilla TD3, without requiring retuning or retraining.



FIGURE 24. TD3-HD performance during reconfiguration: superior adaptation and control precision.

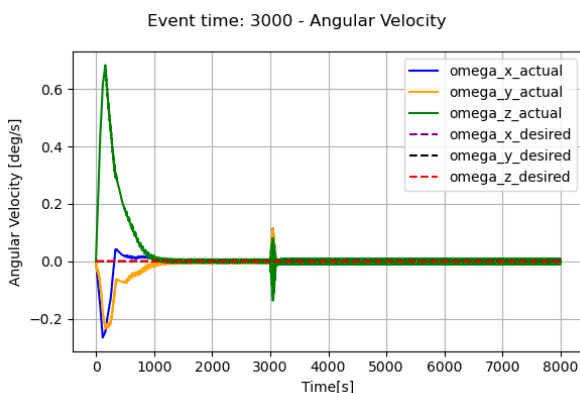


FIGURE 25. TD3-HD detailed metrics during reconfiguration: optimal system response.

C. KEY FINDINGS AND RESEARCH IMPLICATIONS

The experimental investigation yields several significant findings regarding the application of DRL to spacecraft attitude control under uncertain conditions:

1) TRADITIONAL CONTROLLER LIMITATIONS

The conventional PD controller, while effective under nominal operating conditions, demonstrates fundamental

limitations when subjected to system uncertainties. Following the introduction of uncertain conditions at $t = 3000$ seconds, the PD controller exhibits substantial attitude deviations and prolonged angular velocity instabilities, necessitating manual parameter adjustment for stability restoration. These results emphasize the inadequacy of fixed-parameter controllers for autonomous spacecraft operations in uncertain environments.

2) COMPARATIVE DRL ALGORITHM PERFORMANCE

The comparative analysis of PPO, A2C, and TD3 algorithms reveals distinct performance characteristics under uncertain conditions:

PPO demonstrates moderate adaptability with reasonable uncertainty tolerance, though it introduces oscillatory behaviors in torque distribution during system perturbations.

A2C provides flexible torque management capabilities but exhibits slower convergence rates under dynamic system variations.

TD3 achieves superior precision and stability, demonstrating rapid alignment with target orientations, though requiring higher computational resources and exhibiting sensitivity to control over-adjustments.

Overall, TD3 proves most robust among conventional DRL approaches, though it still exhibits limitations compared to the proposed enhanced methodology.

3) ENHANCED TD3-HD PERFORMANCE CHARACTERISTICS

The proposed TD3-HD algorithm demonstrates exceptional adaptability and precision for spacecraft attitude control under uncertain operating conditions. The integration of HER enables efficient learning from sparse reward environments, while Dynamic Weight Clipping optimizes torque adjustments to minimize control fluctuations. These enhancements result in stable attitude alignment and consistently low error levels even under challenging uncertain conditions. The algorithm effectively redistributes torque among available actuators without excessive oscillations, establishing TD3-HD as the most effective autonomous solution for uncertainty-tolerant spacecraft control applications.

4) IMPLICATIONS FOR AUTONOMOUS SPACECRAFT OPERATIONS

This research demonstrates the significant potential of advanced DRL methodologies, particularly the proposed TD3-HD approach, in enhancing spacecraft autonomy through adaptive response to system uncertainties without external intervention. This capability proves essential for mission continuity, where reliability and adaptability are crucial for operational success under unpredictable conditions.

The TD3-HD algorithm's ability to maintain control precision under challenging uncertain scenarios demonstrates its applicability for advanced space missions requiring resilient and adaptive control solutions. These capabilities are

particularly valuable for deep space missions where communication delays preclude real-time ground-based intervention.

VII. CONCLUSION AND FUTURE WORK

This investigation introduced an advanced DRL approach, designated TD3-HD with Dynamic Weight Clipping, to enhance spacecraft attitude control and address system uncertainties in autonomous, fault-tolerant operational scenarios. The TD3-HD methodology combines HER for improved learning from sparse reward environments with DWC for stable actuator utilization, making it exceptionally well-suited for dynamic space environments.

Comprehensive simulations conducted within the Basilisk framework demonstrate that TD3-HD achieves superior attitude error reduction, enhanced angular velocity control, and greater stability under uncertain system conditions compared to traditional PD control and alternative DRL methods (standard TD3, PPO, and A2C). These results highlight TD3-HD's potential as a robust, uncertainty-tolerant, autonomous control solution for spacecraft systems, effectively enabling autonomous uncertainty management and improving system resilience.

Although TD3-HD performs well in simulation, onboard training or adaptation remains challenging due to limited computational, memory, and power resources available on spacecraft. These constraints restrict the feasibility of online learning or large-scale experience replay in orbit, implying that TD3-HD is primarily suited for offline training with fixed policies or limited on-orbit fine-tuning under strict resource and safety constraints.

Future research will extend TD3-HD applications to spacecraft constellation scenarios, focusing on coordinated, uncertainty-tolerant control across multiple spacecraft platforms. Developing distributed DRL frameworks will support autonomous attitude management and adaptability to shared system uncertainties, advancing scalable, resilient operations for multi-spacecraft mission environments. Integration of intrusion/anomaly detection into our TD3-HD recovery framework can be considered as another future work.

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